

Blake Byer

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April 20, 2025

Education

Bachelor of Science in Neuroscience

December 2025

University of Kentucky
Minor in Computer Science
With Honors, 3.95 GPA

Experience

SURE Scholar

June 2024 – August 2024

University of North Carolina at Chapel-Hill

SURE Scholar at the University of North Carolina at Chapel-Hill. I studied rare lysosomal storage disease patients and their COVID-19 outcomes in the National COVID Cohort Collaborative (N3C).

Research Assistant

August 2023 – Present

University of Kentucky

Research Assistant at the University of Kentucky College of Medicine. I research polygenic risk scores for Alzheimer's disease and statistical fine-mapping for GWAS loci prioritization.

Research Assistant

January 2023 – August 2023

University of Kentucky

Research Assistant at the Sports Medicine Research Institute at the University of Kentucky College of Health Sciences. I contributed to a pilot study that tested wireless motion capture and force-plate insoles for gait retraining of military personnel.

Presentations

Office of Graduate Education Symposium

July 2024

University of North Carolina at Chapel-Hill

Lysosomal storage diseases may be protective against severe COVID-19 infection. Clustering analysis shows that lung disease and lysosomal storage diseases further increase infection severity.

[10.5281/zenodo.12971921](https://doi.org/10.5281/zenodo.12971921)

Great Lakes Bioinformatics Conference

May 2024

University of Pittsburgh

Averaging polygenic risk scores derived from multiple genome-wide association studies increases predictive accuracy for Alzheimer's disease.

Selected Publications

1. Averaging polygenic risk scores derived from multiple genome-wide association studies increases predictive accuracy for Alzheimer's disease. Manuscript in preparation.
2. Lysosomal storage diseases and COVID-19. Manuscript in preparation.

Awards

- Great Lakes Bioinformatics Conference Travel Fellowship, International Society for Computational Biology, 2024
- Eagle Scout Award, Boy Scouts of America, 2019
- [Bob and Billie Holcomb Youth Hero Award, Benton Community Foundation, 2017](#)

Memberships

- International Society for Computational Biology, 2024 - Present